

JEE MAIN REVISION MASTER: GEOMETRICAL OPTICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 268 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Geometrical Optics - Core Application	268 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Geometrical Optics - Core Application (Very High Yield — 268 PYQs)

Core Principle: Key problem pattern in Geometrical Optics.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A po..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "A compound microscope is designed with two symmetric biconvex lenses. The objective lens is cut vertically, creating two identical plano-con..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be " [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A point source is placed at a distance of $4R$ in front of spherical interface. The magnitude of the magnification of point source image is ____.

- (A) 1.66 **[Correct]**
- (B) 2.33
- (C) 2.66
- (D) 1.33

Correct Answer: A

Solution Analysis: When light travels from one medium to another through a curved surface, the positions of the object (u) and the image (v) are related by the formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ Where n_1 is the refractive index of the medium where the object is placed, n_2 is the refractive index of the medium where light enters, R is the radius of curvature. The refractive index of medium in which object is $n_1 = 1$, and that for the image $n_2 = 1.4$. The object distance is $u = -4R$ (as per sign convention), and radius is $+R$ (convex surface). $\frac{1.4}{v} - \frac{1}{-4R} = \frac{1.4 - 1}{R}$